DEGREE: MSc in Artificial Intelligence

Module: Introduction to Artificial Intelligence

Assignment Title: Predictive Maintenance in Smart Manufacturing: Evaluating AI Models for Equipment Failure Prediction

Assignment Type: Essay

Word Limit: 1000-1500 words

Weighting: 50%

Issue Date: 24/04/2026

Submission Date: 17/06/2026

Feedback Date: 01/07/2026

Issued by: Dr. Abdelaziz Triki

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You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration		
Word count		
Use of proof-reader/proof-reading service (see guidance contained in Academic Misconduct Regulations on appropriate use)	**YES**	**NO**

If 'yes', please specify which service (e.g. Grammarly, Studiosity etc.)		
I confirm that I submit this work as my own work and that I have cited all sources I have used, and I understand that using sources without citing them correctly may be considered as Academic Misconduct.	**YES**	**NO**
I confirm that I have followed guidance on the acceptable use of AI tools for this assignment where such guidance has been issued by my tutors.	**YES**	**NO**
Where use of appropriately cited use of AI tools is permitted, I confirm that I have cited in accordance with the UCA Harvard Referencing Standard.	**YES**	**NO**

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The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

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Signature (Student): ______________________ Date: ___________________

Introduction

Industrial manufacturing companies increasingly rely on predictive maintenance strategies to minimize unplanned downtime and optimize asset performance. In the era of Industry 4.0, interconnected sensors embedded within machines collect vibration, temperature, acoustic and current signals. Several AI models analyze this data to forecast potential failures and schedule maintenance proactively. Recent studies emphasize that predictive maintenance is a critical pillar for efficient operations and can revolutionize maintenance practices; however, challenges remain around data quality, model interpretability and scalability. AI models serve as the cornerstone of predictive maintenance by automating analysis of vast sensor datasets and enabling the proactive identification of equipment faults. Yet the reliability of these systems depends on high-quality labelled data, transparent models and robust governance frameworks.

Learning Outcomes:

LO1. Demonstrate the ability to design and execute AI experiments, showing innovative problem-solving skills and adapt AI techniques to new challenges

LO2. Conduct in-depth research on AI trends and technologies, analyze the latest advancements, and assess potential impact on society

LO3. Engage in a collaborative real problem-based projects using modern AI tools

Assessment Criteria: Weighting 50%
1000-1500 words

Problem Statement / Case Study

You are part of a data science team at a smart manufacturing company that operates a network of industrial pumps, motors and conveyor systems. The company experiences unexpected equipment failures that halt production and incur significant costs. Each machine is instrumented with sensors that record vibration, temperature, pressure and acoustic signatures at high frequency. Management wants to develop a predictive maintenance solution that uses AI models to estimate the remaining useful life of each component and to flag anomalies before breakdowns occur. Your task is to investigate state-of-the-art AI approaches for predictive maintenance, compare at least two suitable models and outline an experiment design to evaluate them on available sensor data.

Tasks

Task 1: Literature Review (2021–2025)

Conduct a systematic literature review of recent peer-reviewed research on AI-driven predictive maintenance in industrial settings. Your review should:

- Identify and synthesise a **minimum of 10 academic sources published between 2021 and 2025**, sourced from reputable databases such as IEEE Xplore, Scopus, SpringerLink, or ACM Digital Library.
- Critically evaluate the state of the art in AI-based predictive maintenance, highlighting dominant methodologies, benchmark datasets (e.g., NASA CMAPSS, PRONOSTIA bearing dataset), and key performance outcomes reported in the literature.
- Identify **research gaps or limitations** acknowledged in existing studies (e.g., lack of labelled failure data, poor generalisation across machine types, real-time deployment challenges) and discuss how your proposed approach addresses one or more of them.
- Avoid descriptive summaries. Your review must demonstrate **critical engagement**, comparing and contrasting findings across sources.

Task 2: AI Model Selection and Justification

Choose **at least two AI models** suitable for sensor-based fault prediction and Remaining Useful Life (RUL) estimation. Your selection must be academically justified, not arbitrary (e.g., Random Forest vs. Gradient Boosted Trees; Long Short-Term Memory (LSTM) networks vs. Temporal Convolutional Networks).

Task 3: Data Requirements and Preprocessing

- Discuss data requirements and typical preprocessing steps, including handling imbalance, feature extraction from time-series signals, and splitting data into training and test sets.

Task 4: Experimental Design

- Propose an experimental design outlining evaluation metrics (such as precision, recall, F1 score, remaining useful life estimation error), cross-validation strategies and comparison criteria.

Task 5: Critical Analysis: Interpretability, Scalability, and Deployment

Critically evaluate your chosen models beyond raw performance, addressing the following three dimensions:

- **Interpretability**: Assess how explainable each model is in a real-world industrial context. Compare the inherent transparency of tree-based models against the black-box nature of deep learning approaches, and discuss how post-hoc tools such as SHAP values or LIME could support maintenance decision-making in safety-critical environments.
- **Scalability**: Analyse how each model handles growing data volumes, additional machines, and the need for periodic retraining. Consider both computational cost (memory, training time) and operational overhead (model versioning and monitoring pipelines).

- **Deployment**: Discuss the practical challenges of deploying your models in a live manufacturing environment, including edge vs. cloud trade-offs, real-time latency requirements, integration with existing industrial systems (e.g., SCADA/MES), and strategies for detecting model drift over time.

Submission Guidelines

- Prepare your essay using the **BSBI assignment template** available on Canvas.
- Ensure the structure is clear, well-organised, and visually well-presented.
- Include tables, figures, or diagrams where appropriate (label and reference them clearly).
- Minimum **10 academic sources (2021–2025)** using **Harvard referencing**.
- Submit **one PDF file** via Canvas before the deadline.

Students are encouraged to explore additional reputable sources in the digital library beyond the list provided to strengthen their individual research and critical analysis skills. The unit lecturer can help with recommending additional literature.

The criteria for successfully passing the units with two assessment components require passing both components successfully. Passing marks for BSc and MA/MSc levels are different and are stated at the end of the assignment brief. Please find the examples below:

· **Example 1** (e.g., pass mark: 50)
· Component 1 (50% weighting) - 55
· Component 2 (50% weighting) - 55
· Final average: 55
· Result: **Pass (both components passed and therefore the unit is passed)**

· **Example 2** (e.g., pass mark: 50)
· Component 1 (50% weighting) – 80
· Component 2 (50% weighting) – 30
· Final average: 55
· Result: **Fail (because although the average is above 50, one of the components is a failure**)

GRADING DESCRIPTORS: LEVEL 7

EXPERIMENTATION & INNOVATION								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Deals with complex issues both systematically and creatively demonstrating self-direction and originality in tackling and solving problems	Little to no ability to use techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions.	Low utilisation of established techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions, but with limitations in techniques or approach.	Limited research or advanced scholarship to their area of study by using a range of information and established and advanced techniques	Competent understanding of solving problems, through own research or advanced scholarship displaying a comprehensive understanding of established and advanced techniques	Good understanding of solving problems through own research and advanced scholarship critically selecting and displaying a comprehensive understanding of established and advanced techniques.	Very Good problem-solving skills displaying a comprehensive understanding of techniques applicable to their own research or advanced scholarship	Excellent range of extremely well-developed problem-solving displaying an understanding of techniques applicable to their own research or advanced scholarship beyond which is taught.	Exceptional problem-solving skills with sophisticated evaluation and application of a wide range of advanced information and techniques to undertake projects.
Comprehensive understanding of techniques applicable to their own research or advanced scholarship	Little to no understanding of techniques applicable to their own research or advanced scholarship or their limitations and ambiguities.	Low understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Limited understanding of key techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Competent understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities	Good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Very good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Excellent understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.	Exceptional understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.

GRADING DESCRIPTORS: LEVEL 7

RESEARCH & ANALYSIS								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Systematic understanding of knowledge, and a critical awareness of current problems and/or new insights, much of which is at, or informed by, the forefront of their academic discipline, field of study or area of professional practice	Little to no knowledge of the subject with limited breadth or depth or deficiencies in major areas or currency.	Low knowledge of the subject lacking coherence, breadth, or detail with only some reference to ideas or arguments at the forefront of any part of the subject.	Limited knowledge to deal with terminology, facts and concepts some of which is informed by the forefront of defined areas of the subject.	Competent knowledge of ideas or arguments at the forefront of any part of the subject sufficient to deal with current issues in the discipline, generally more descriptive than critical or analytical.	Good knowledge of ideas or arguments at the forefront of any part of the subject showing a clear, critical insight into the discipline as whole and current issues/problems.	Very good knowledge of ideas or arguments at the forefront of the subject some of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Excellent knowledge of ideas or arguments at the forefront of the subject many of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Exceptional knowledge of ideas or arguments at the forefront of the subject most of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.
Conceptual understanding that enables the student to display originality in the application of knowledge	Little to no conceptual understanding or argument and a focus on descriptive explanations which do not comment on arguments of others or alternative views.	Low conceptual understanding and arguments are weak or poorly constructed, and the work does not critically evaluate the arguments of others or consider alternative views.	Limited conceptual understanding and argument construction with critical evaluation of alternative views or comment on advanced scholarship.	Competent conceptual understanding and argument construction with critical evaluation of a range of views and consistent engagement with advanced scholarship.	Good conceptual understanding which critically evaluate and synthesise other views and information with a thoughtful interpretation of advanced scholarship.	Very good conceptual understanding which systematically synthesises a wide range of views with a critical insight into advanced scholarship.	Excellent conceptual understanding which critically apply a wide range of views through a perceptive use of advanced scholarship.	Exceptional conceptual understanding of publishable quality with systematic engagement and usage of advanced scholarship.

GRADING DESCRIPTORS: LEVEL 7

ENGAGING WITH PRACTICE								
	FAIL			PASS				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Practical understanding of how established techniques of research and enquiry are used to create and interpret knowledge in the discipline	Little to no evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Low evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Limited background investigation, analysis, research, enquiry , ethical awareness, and/or study using established techniques, with the ability to extract relevant points.	Competent investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and can critically appraise and use academic sources.	Good background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and possesses a well-developed ability to critically appraise a wide range of sources.	Very good, independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Excellent independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study well beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Exceptional investigation, analysis, research, enquiry, ethical awareness, and/or study which demonstrates carefully considered depth and breadth and critically synthesises this to advance the work and/or direct arguments.
Originality in the application of knowledge	Little to no technical, creative or artistic skills related to their area of study.	Low technical, creative or artistic skills related to their area of study.	Limited technical, creative or artistic skills required for area of study.	Competent technical, creative or artistic skills required for area of study.	Good technical, creative or artistic skills required for area of study.	Very good range of technical, creative or artistic skills.	Excellent range of technical, creative or artistic skills	Exceptional range of technical, creative or artistic skills
Independently advance your own knowledge and understanding, and to develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.

GRADING DESCRIPTORS: LEVEL 7

REALISATION & COMMUNICATION								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Communicate information, ideas, problems and solutions to both specialist and non-specialist audiences.	Little to no clarity in the communication of ideas, problems and solutions to audiences.	Low clarity in the communication of ideas, problems and solutions to audiences.	Limited clarity in the communication of ideas, problems and solutions to audiences.	Competent communication of ideas, problems and solutions to audiences.	Good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Very good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Excellent communication of ideas, problems and solutions to audiences in a range of means / media.	Exceptional communication of ideas, problems and solutions to audiences in a range of means / media.

GRADING DESCRIPTORS: LEVEL 7

PERSONAL & PROFESSIONAL CONNECTIVITY								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Independently advance your own knowledge and understanding, and develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.
Qualities and transferable skills necessary for employment requiring: (a) the exercise of initiative, ethical and personal responsibility (b) decision-making in complex and unpredictable contexts	Little to no ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Low ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Limited ability to manage learning and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Competent ability to manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Very good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations.	Excellent ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Exceptional ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations
	Little to no use of appropriate terminology, limited vocabulary and many errors in spelling, grammar and syntax.	Low use of appropriate terminology, with many errors in spelling, vocabulary and syntax.	Limited expression, style and appropriate vocabulary with errors in spelling, grammar and syntax which affect understanding.	Competent expression, style, and appropriate vocabulary with some errors in spelling, grammar and syntax which do not affect understanding.	Good expression, style and appropriate vocabulary with some errors in spelling, grammar and syntax.	Very good expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Excellent expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Exceptional expression, style and appropriate vocabulary with no errors in spelling, grammar and syntax.

GRADING DESCRIPTORS: LEVEL 7

	Little to no evidence of basic numeracy or digital literacy, hardware and software skills	Low evidence of basic numeracy or digital literacy, hardware and software skills competency.	Limited evidence of numeracy or digital literacy, hardware and software skills competency.	Adequate evidence of numeracy or digital literacy, hardware and software skills competency.	Good evidence of numeracy or digital literacy, hardware and software skills competency.	Very good evidence of numeracy or digital literacy, hardware and software skills	Excellent evidence of numeracy or digital literacy, hardware and software skills competency.	Exceptional evidence of numeracy or digital literacy, hardware and software skills competency.

	competency.					competency.		
	Does not demonstrate achievement of professional competence when assessed against the requirements of a professional, statutory or regulatory body (PSRB).			The student has demonstrated achievement of professional competence when assessed against the requirements of a PSRB.				
	Inaccurate use of terminology with limited vocabulary and many errors in spelling, grammar and syntax. Inaccurate terminology, with many errors in spelling, vocabulary and syntax.			The student has adhered to the appropriate rules and/or conventions set by regulators or the industry.				